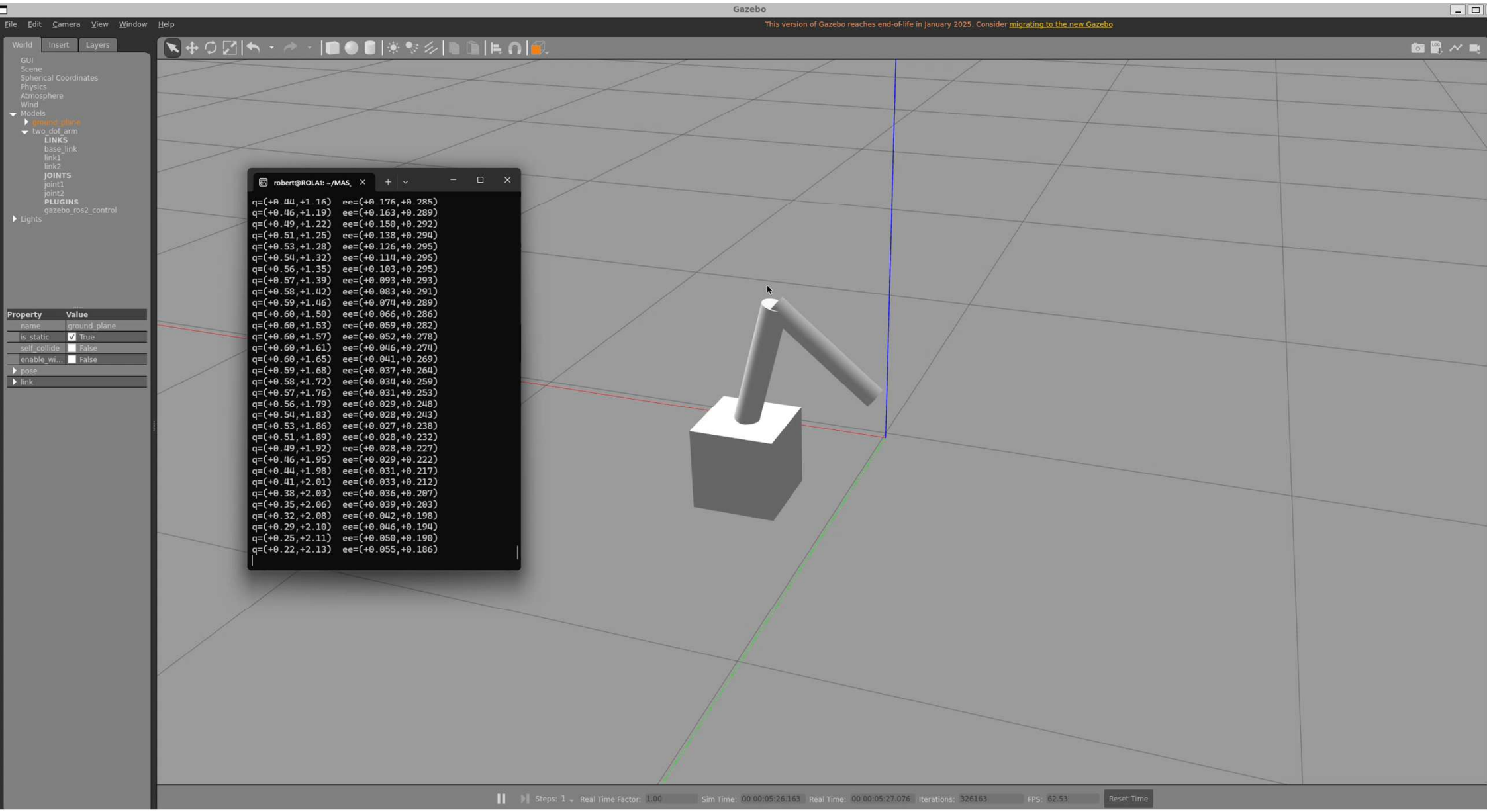
Mechanical **A**erial **S**ystems
(MAS)

F26, SDU Dronecenter

Robert Ladig – rola@sdu.dk


Gazebo
File Edit Camera View Window Help
This version of Gazebo reaches end-of-life in January 2025. Consider migrating to the new Gazebo
World Insert Layers
GUI
Scene
Spherical Coordinates
Physics
Atmosphere
Wind
Models
ground_plane
two_dof_arm
LINKS
base_link
link1
link2
JOINTS
joint1
joint2
PLUGINS
gazebo_ros2_control
Lights
Property Value
name ground_plane
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self_collide False
enable_wi... False
pose
link
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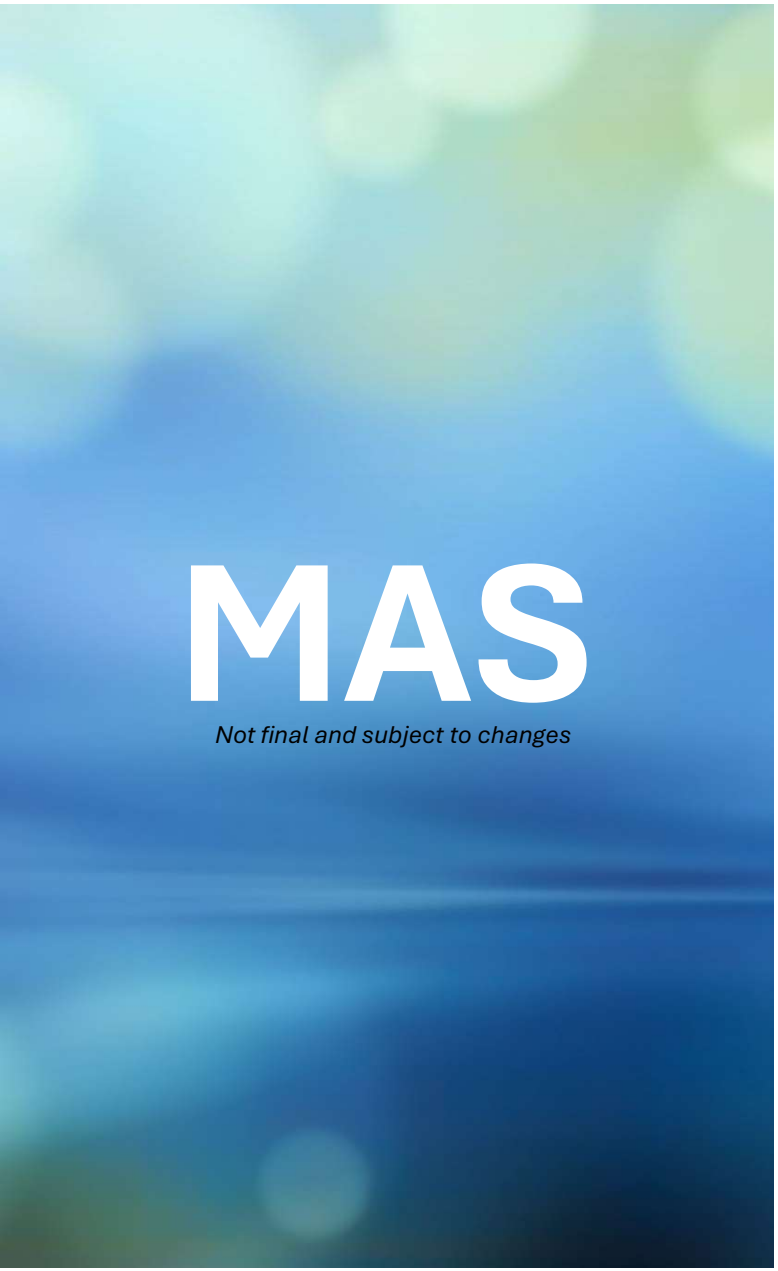


Not final and subject to changes

Unit 1 – Course goal, Intro, Housekeeping, Aerial Manipulation Intro, Simulator Setup, ROS2 (refresher ?!)
Unit 2 - Manipulator types, FK (refresher), IK (refresher), URDF (refresher?!), Simulator work

Unit 3 – Jacobians and coupled kinematics, Simulator work
Unit 4 – PID control (refresher!), Simulator implementation of coupled kinematics for SDU UAS system
Unit 5 – Material handover, Material check and prototyping of own aerial manipulator.

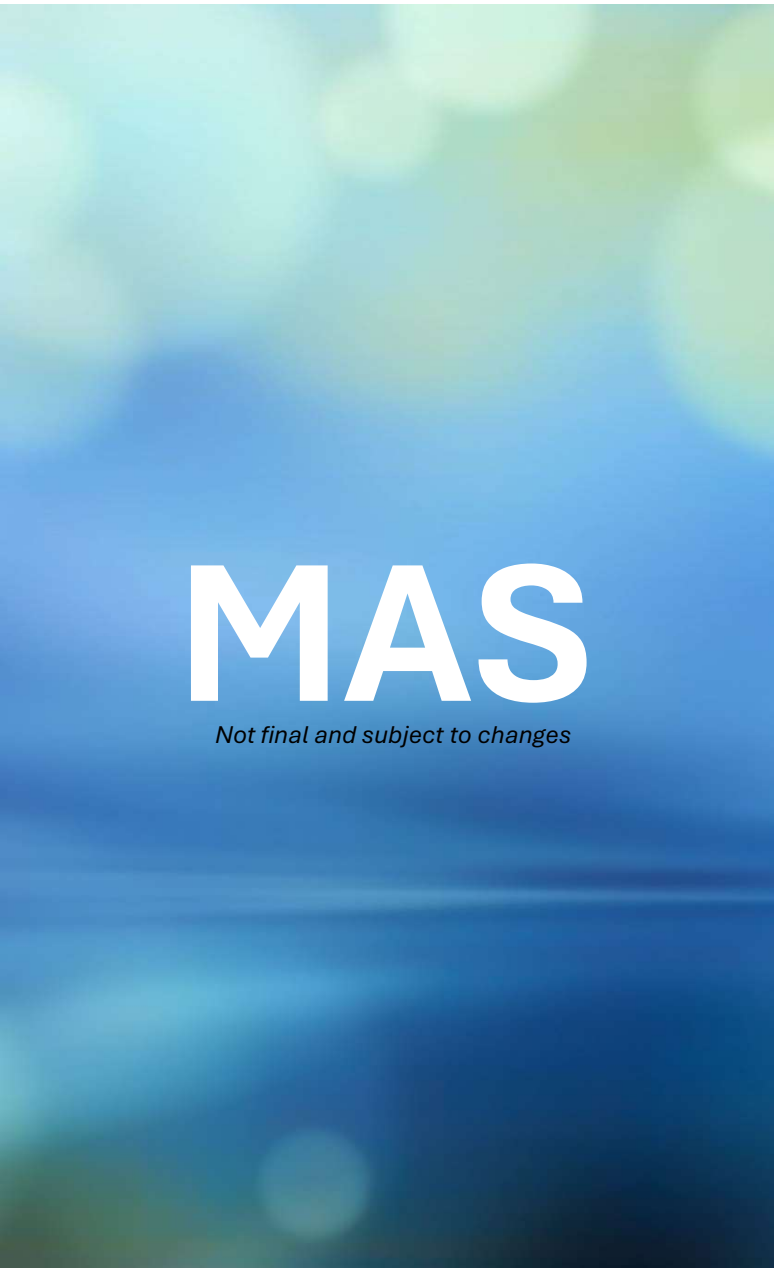
Unit 6 - Group presentation of concept, CAD and refined Unit 4 simulator

----------Flight lab is available to use and drones are RTF----------

Unit 7 – Intro to Flightlab, EIT playground, Offboard control test @ Flightlab
Unit 8 - Bench prototype of Aerial Manipulator
Unit 9 - Airframe integration
Unit 10 – Airframe integration, Flight test I, (opt.) Sensing and control integration

Unit 11 – Final group presentation, Flight test II
Unit 12 – Flight demo , Project report due

Not final and subject to changes



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Manipulator types for aerial manipulation

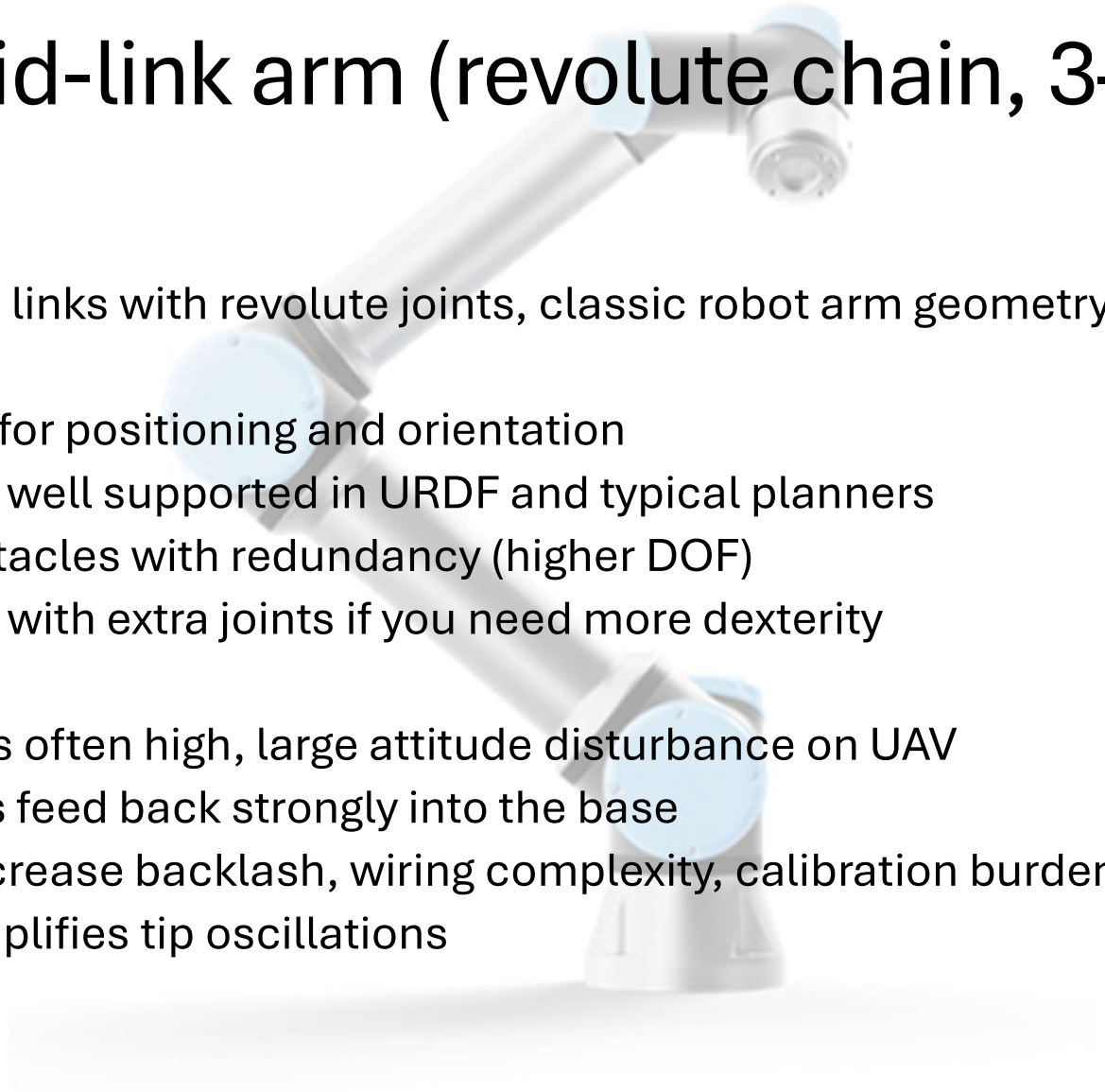
What we mean by “manipulator type”

- The arm’s kinematic structure (links + joints + actuation placement)
- How it shapes workspace, stiffness, mass distribution, and control effort

What we want to compare

- Workspace shape and usable reach
- Moving mass and inertia (how much the drone “feels” the arm motion)
- Stiffness vs compliance (interaction safety and precision)
- Control and calibration complexity
- Mechanical robustness and maintainability

Serial rigid-link arm (revolute chain, 3–7 DOF)



What it is

- A chain of rigid links with revolute joints, classic robot arm geometry

Advantages

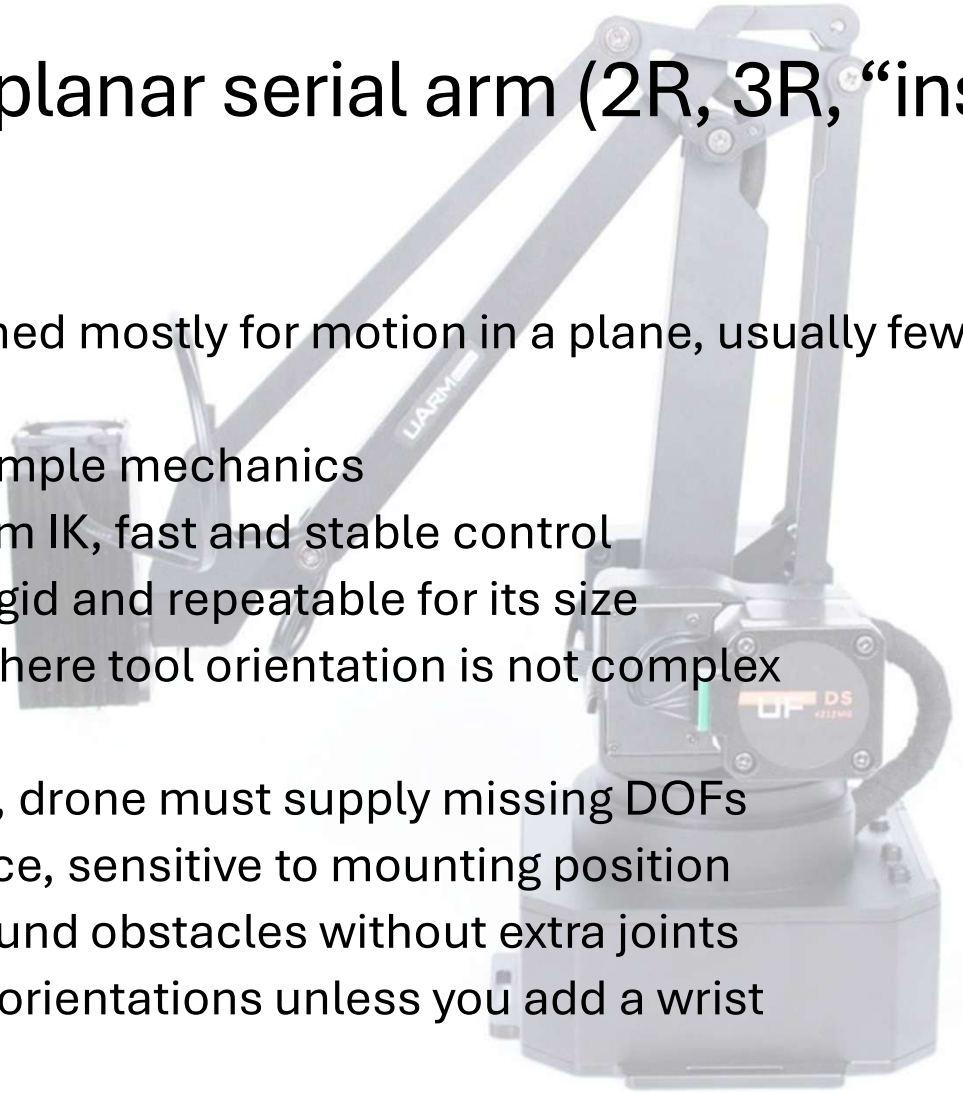
- Most versatile for positioning and orientation
- Intuitive FK/IK, well supported in URDF and typical planners
- Can avoid obstacles with redundancy (higher DOF)
- Easy to extend with extra joints if you need more dexterity

Disadvantages

- Moving mass is often high, large attitude disturbance on UAV
- Contact forces feed back strongly into the base
- Many joints increase backlash, wiring complexity, calibration burden
- Long reach amplifies tip oscillations

UARM
DS

Lightweight planar serial arm (2R, 3R, “inspection arm”)



What it is

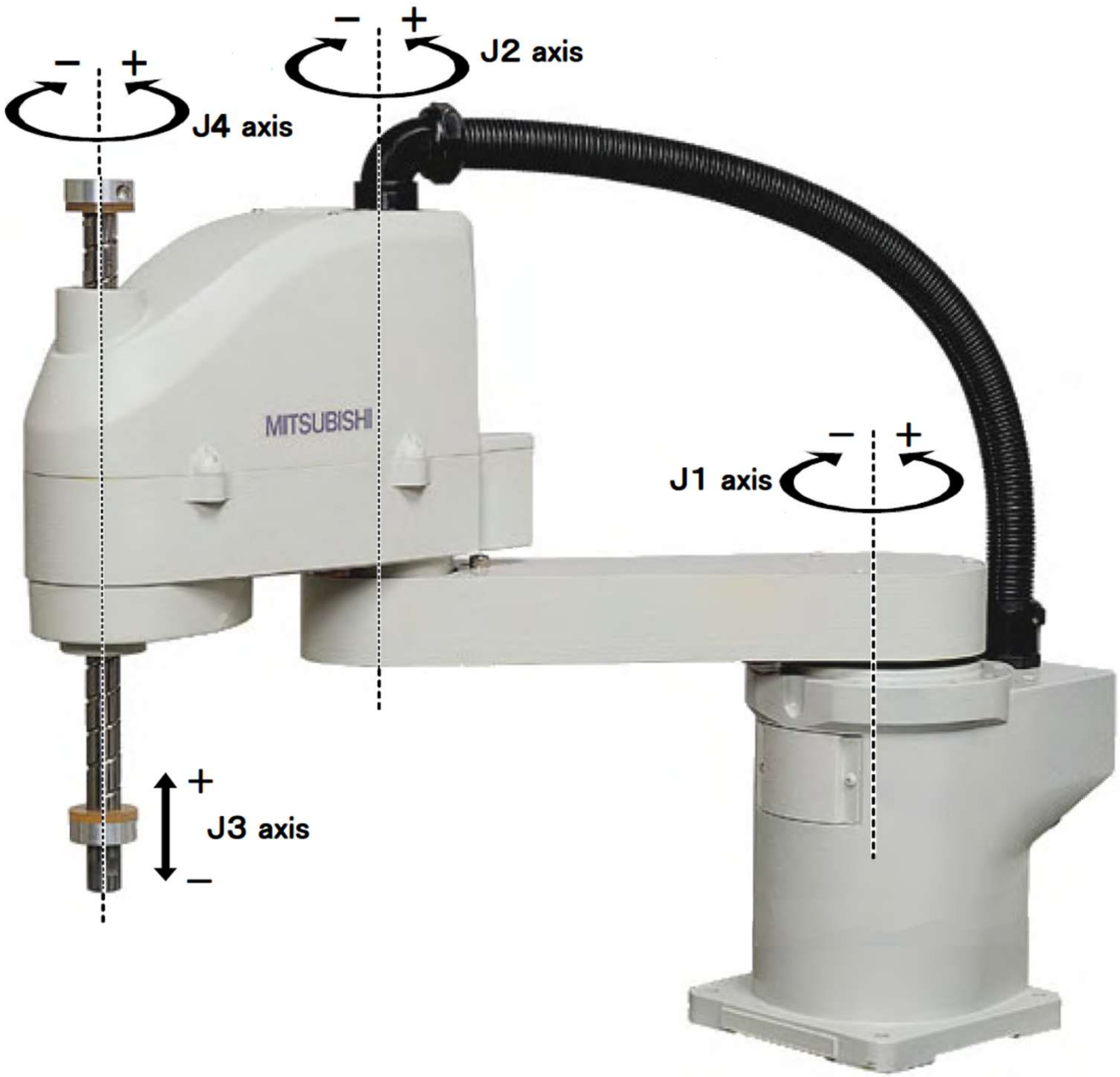
- Serial arm designed mostly for motion in a plane, usually fewer joints

Advantages

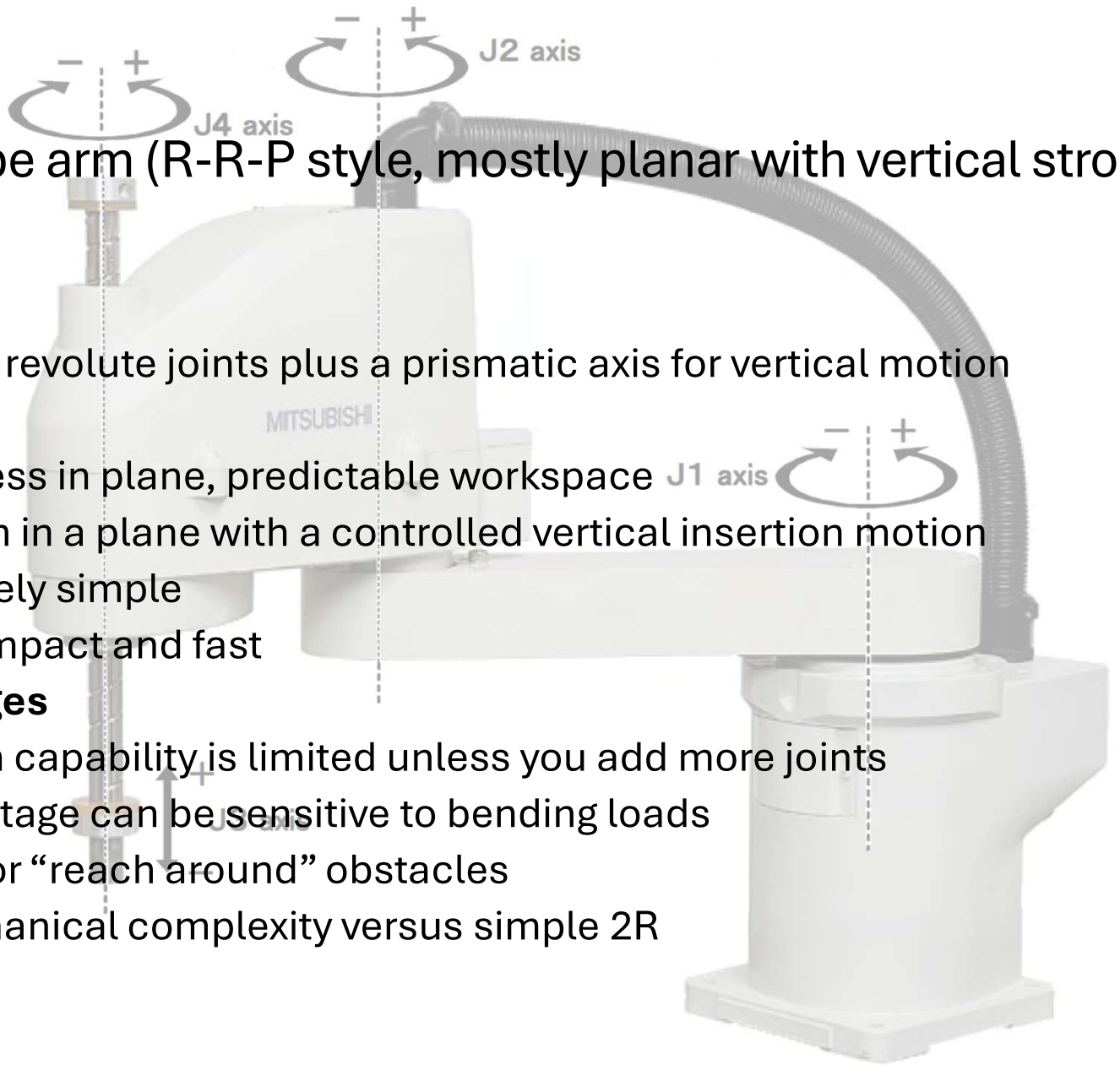
- Low mass and simple mechanics
- Often closed-form IK, fast and stable control
- Easier to make rigid and repeatable for its size
- Good for tasks where tool orientation is not complex

Disadvantages

- Limited dexterity, drone must supply missing DOFs
- Narrow workspace, sensitive to mounting position
- Hard to work around obstacles without extra joints
- Poor at complex orientations unless you add a wrist


J2 axis
J4 axis
J1 axis
J3 axis
MITSUBISHI

SCARA-type arm (R-R-P style, mostly planar with vertical stroke)



What it is

- Two planar revolute joints plus a prismatic axis for vertical motion

Advantages

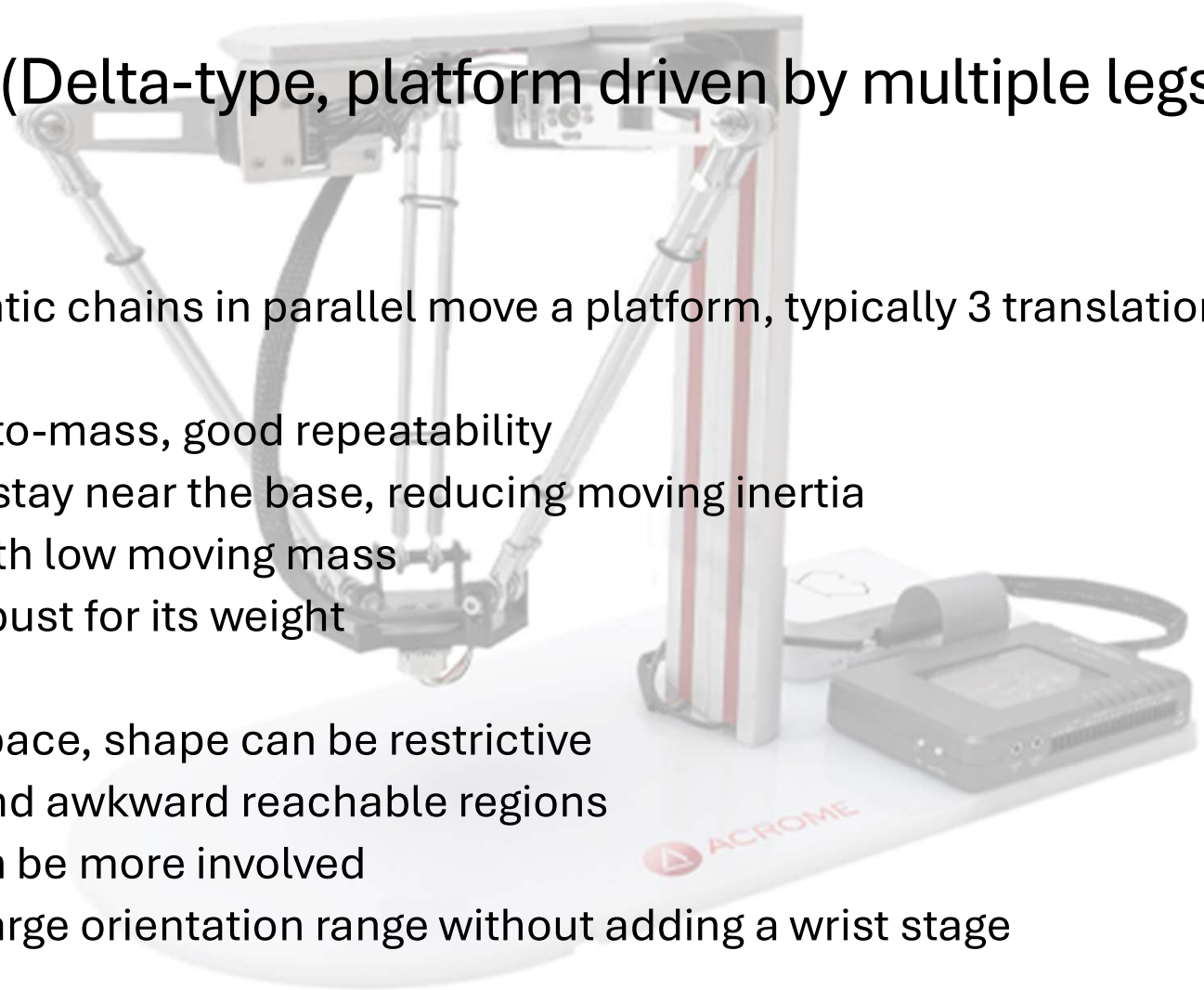
- High stiffness in plane, predictable workspace
- Good reach in a plane with a controlled vertical insertion motion
- IK is relatively simple
- Can be compact and fast

Disadvantages

- Orientation capability is limited unless you add more joints
- Prismatic stage can be sensitive to bending loads
- Not ideal for “reach around” obstacles
- Adds mechanical complexity versus simple 2R

ACROME

Parallel arm (Delta-type, platform driven by multiple legs)



What it is

- Several kinematic chains in parallel move a platform, typically 3 translational DOF

Advantages

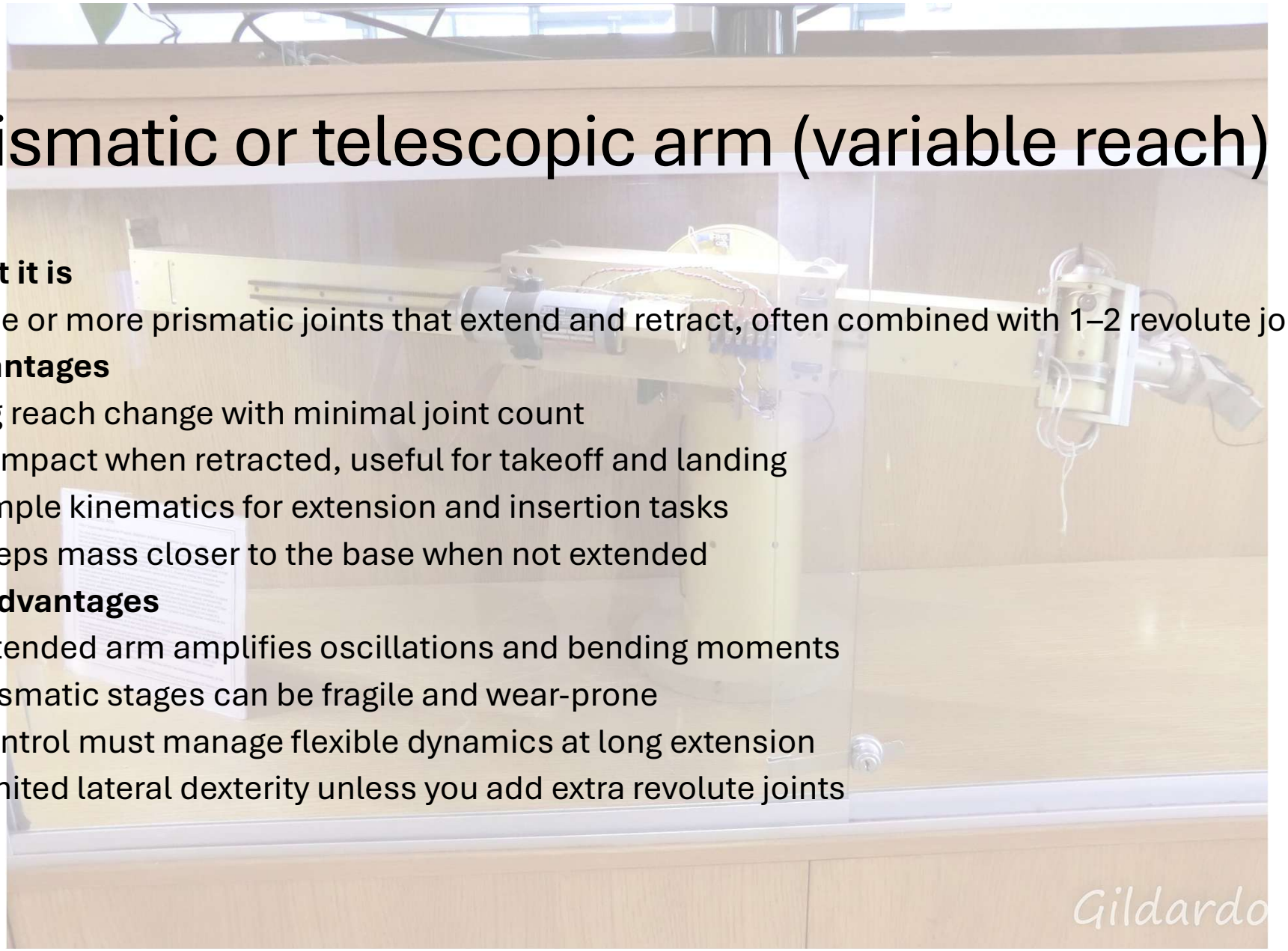
- High stiffness-to-mass, good repeatability
- Actuators can stay near the base, reducing moving inertia
- Fast motion with low moving mass
- Structurally robust for its weight

Disadvantages

- Limited workspace, shape can be restrictive
- Singularities and awkward reachable regions
- Calibration can be more involved
- Harder to get large orientation range without adding a wrist stage

The Stanford Arm
Gildardo

Prismatic or telescopic arm (variable reach)



What it is

- One or more prismatic joints that extend and retract, often combined with 1–2 revolute joints

Advantages

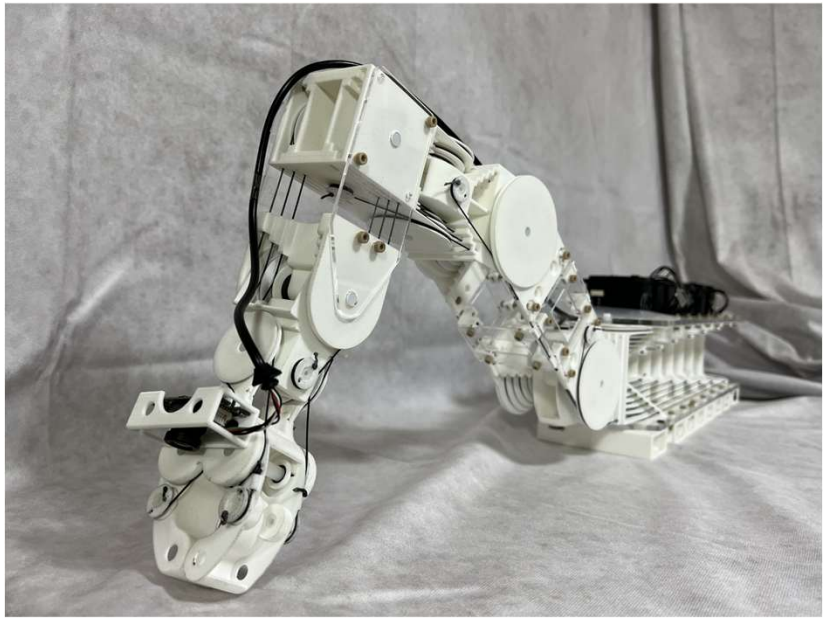
- Big reach change with minimal joint count
- Compact when retracted, useful for takeoff and landing
- Simple kinematics for extension and insertion tasks
- Keeps mass closer to the base when not extended

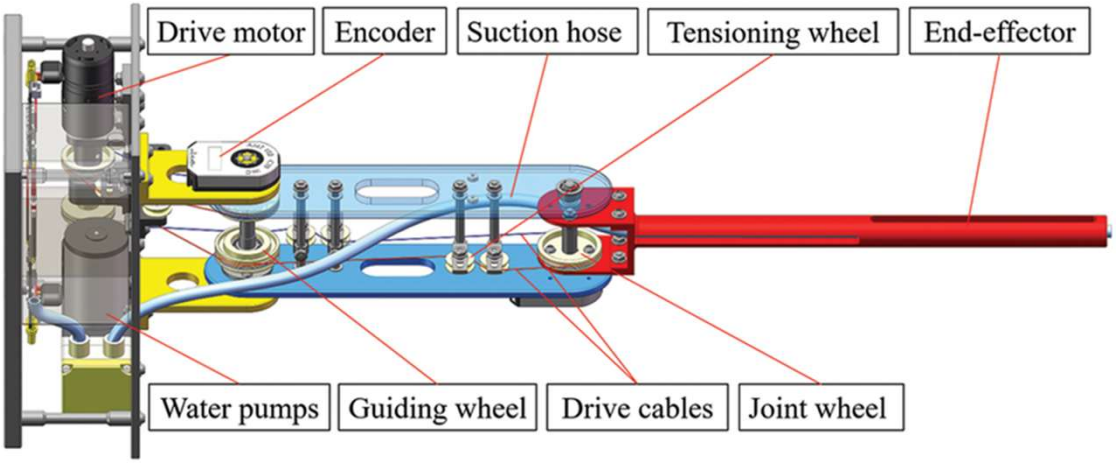
Disadvantages

- Extended arm amplifies oscillations and bending moments
- Prismatic stages can be fragile and wear-prone
- Control must manage flexible dynamics at long extension
- Limited lateral dexterity unless you add extra revolute joints

Drain mouth

Aerial manipulator





Tendon or cable-driven serial arm (remote actuation)

What it is

- Joints actuated via tendons, motors placed near the base to reduce distal mass

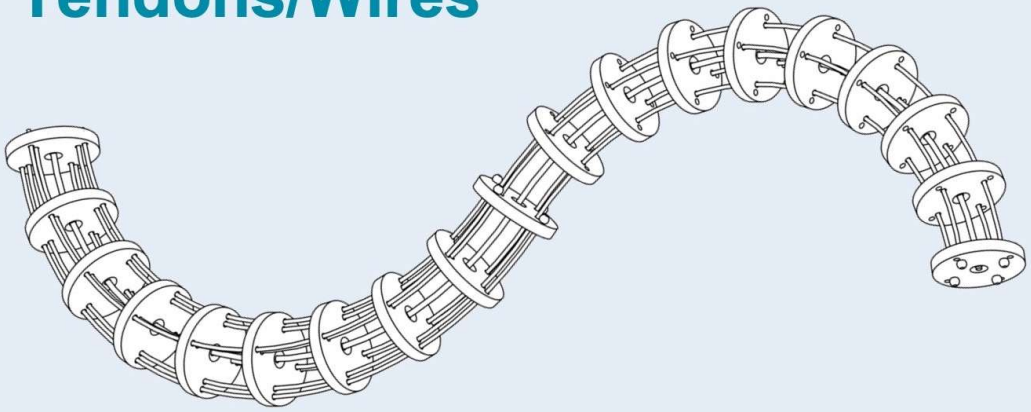
Advantages

- Very low moving mass, reduced disturbance to the UAV
- Can route actuation around geometry constraints
- Potentially safer contact due to inherent compliance
- Slim and lightweight link design possible

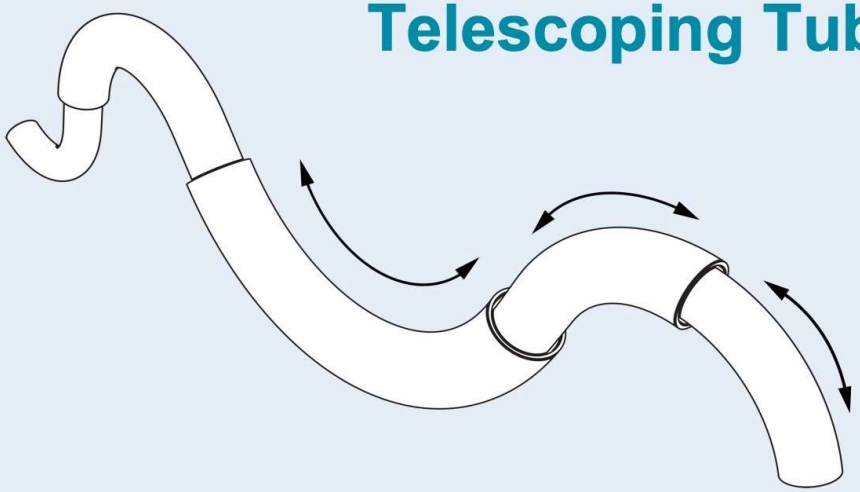
Disadvantages

- Hysteresis, stretch, and backlash complicate control
- Needs careful tensioning and maintenance
- Harder to get accurate torque control without good sensing
- Modeling is less “ideal” than rigid direct-drive joints

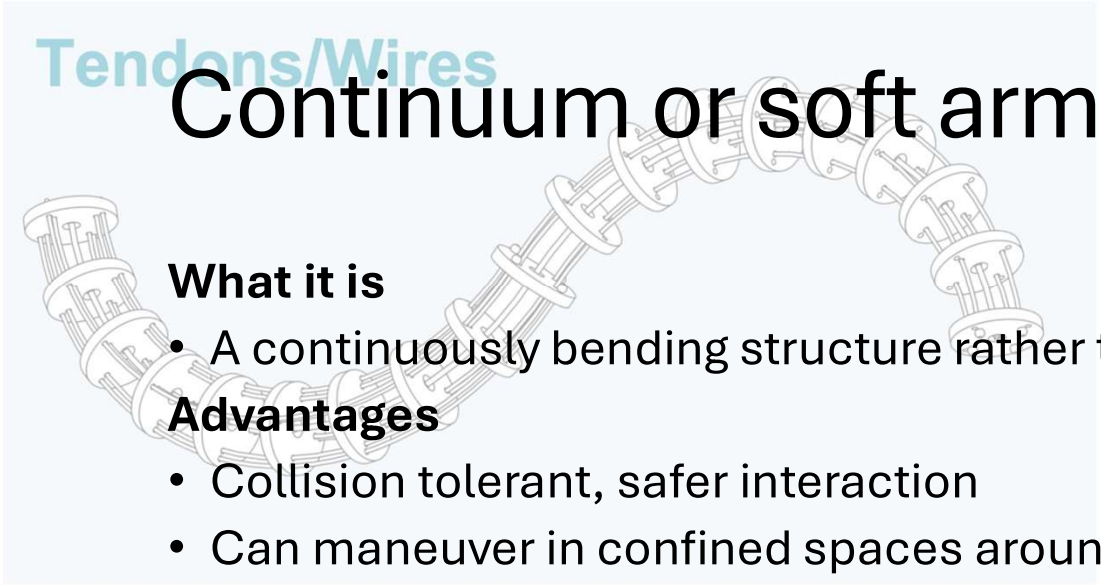
Tendons/Wires

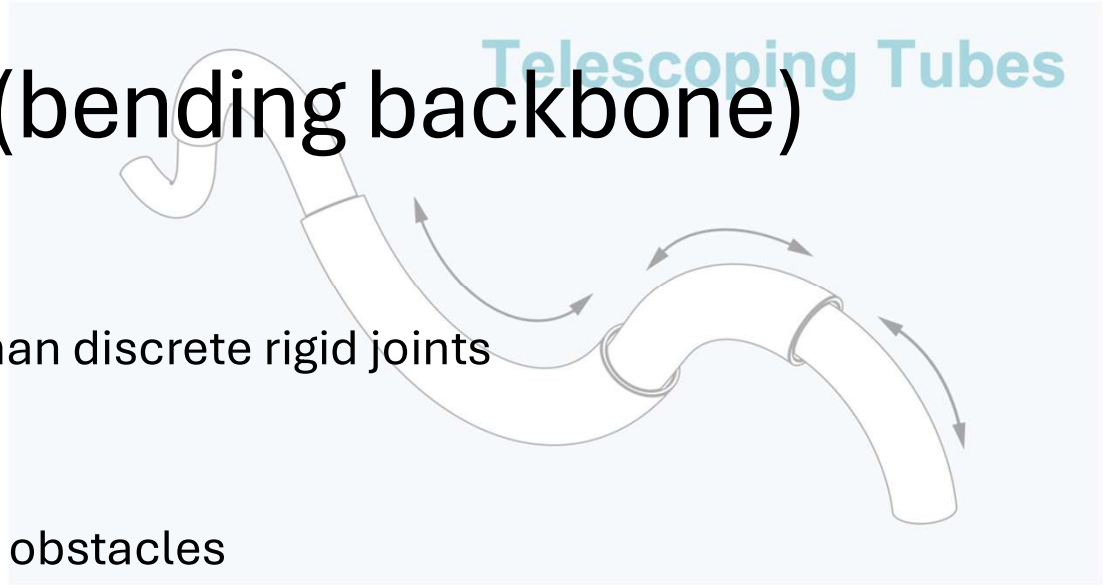


Telescoping Tubes



Continuum or soft arm (bending backbone)





What it is

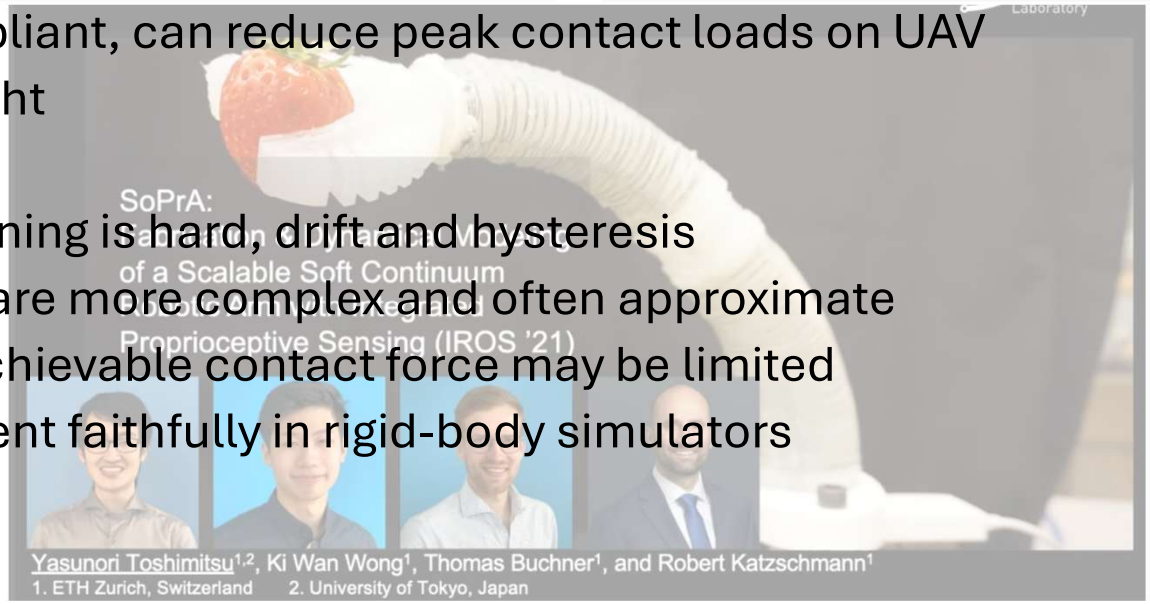
- A continuously bending structure rather than discrete rigid joints

Advantages

- Collision tolerant, safer interaction
- Can maneuver in confined spaces around obstacles
- Naturally compliant, can reduce peak contact loads on UAV
- Often lightweight

Disadvantages

- Precise positioning is hard, drift and hysteresis
- IK and control are more complex and often approximate
- Payload and achievable contact force may be limited
- Hard to represent faithfully in rigid-body simulators



Tangent: Folding arm (stowable geometry)

What it is

- Arm that folds for transport and unfolds for operation, sometimes with lockable joints

Advantages

- Compact for takeoff, landing, and safety
- Can change workspace shape for different phases of the mission
- Protects links and joints during transit
- Helps with packaging on small UAV frames

Disadvantages

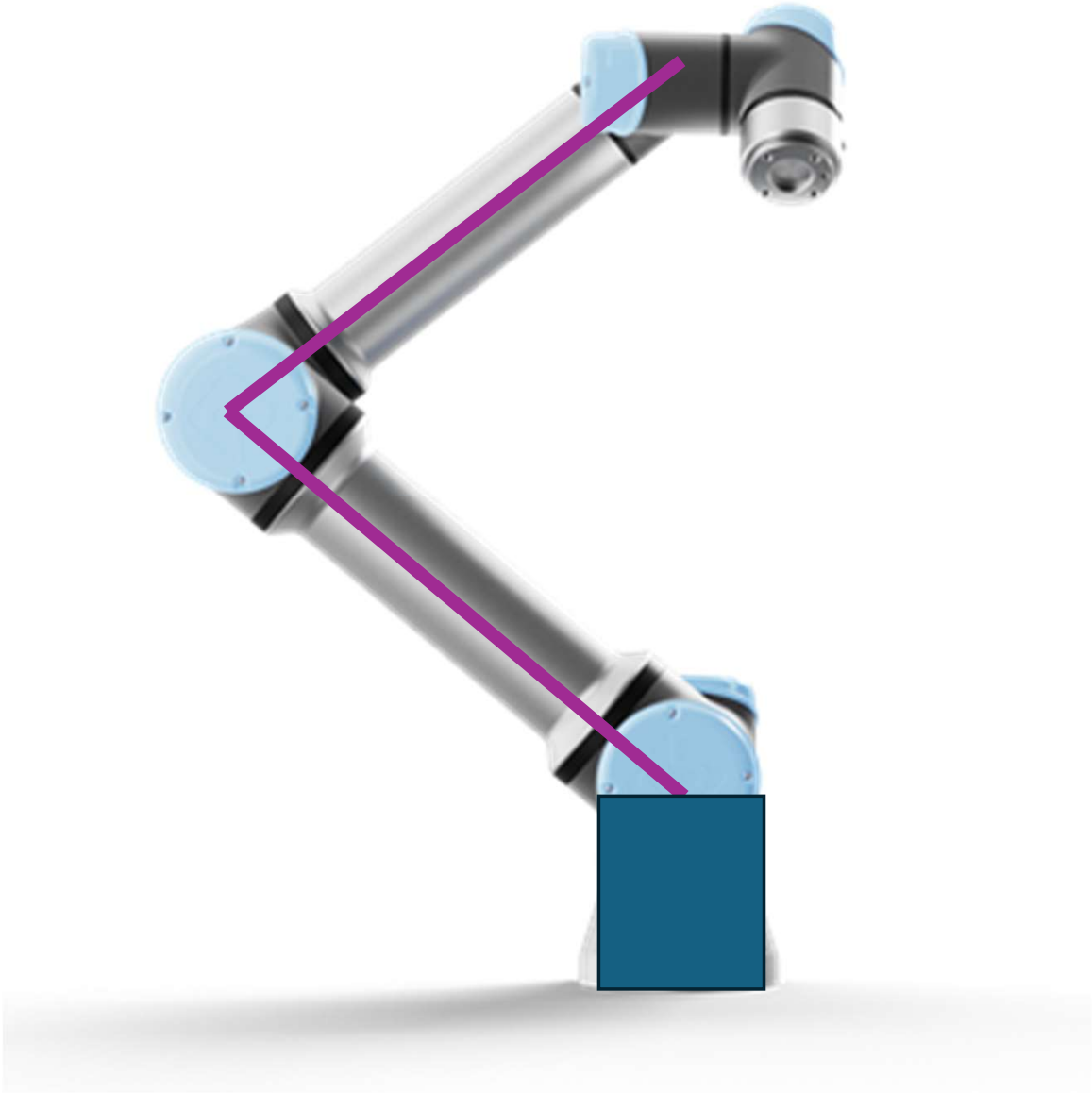
- More mechanical parts, more failure modes
- Joint locks introduce play if not designed well
- Configuration changes complicate planning and calibration
- Added mass from hinges, locks, or deployment mechanisms

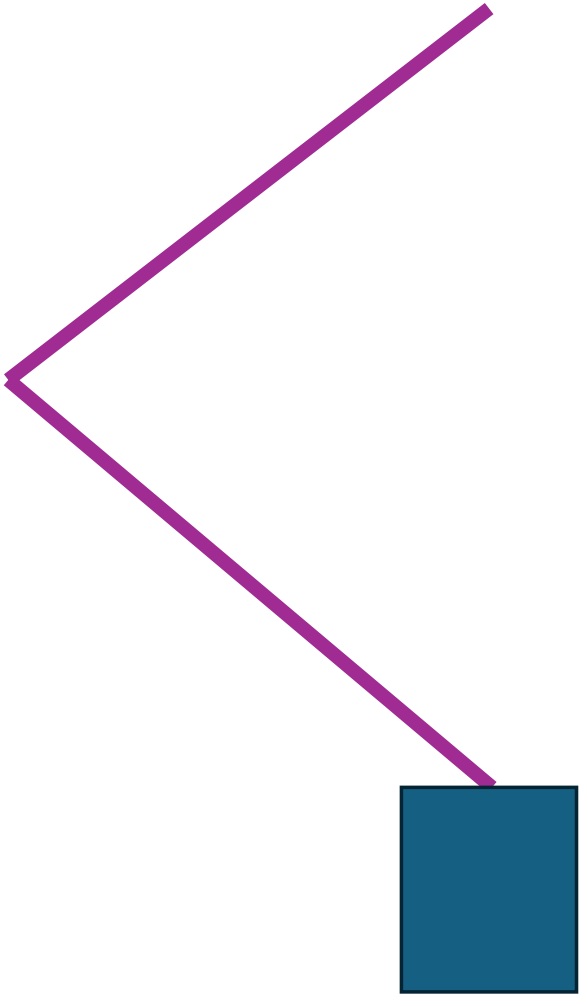
Saros Z70

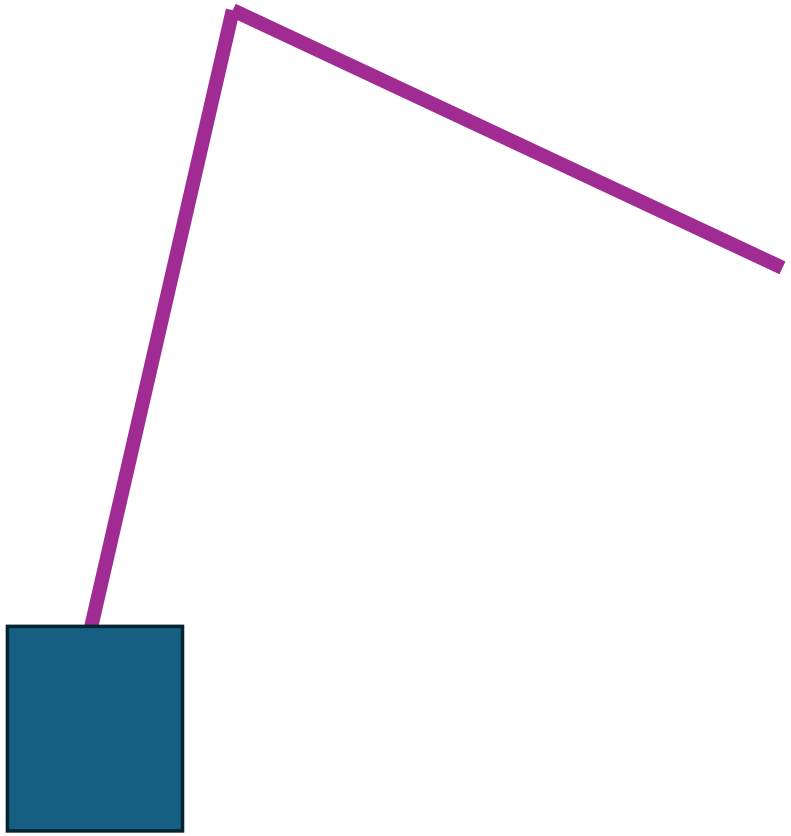
One Arm. Infinite Possibilities.

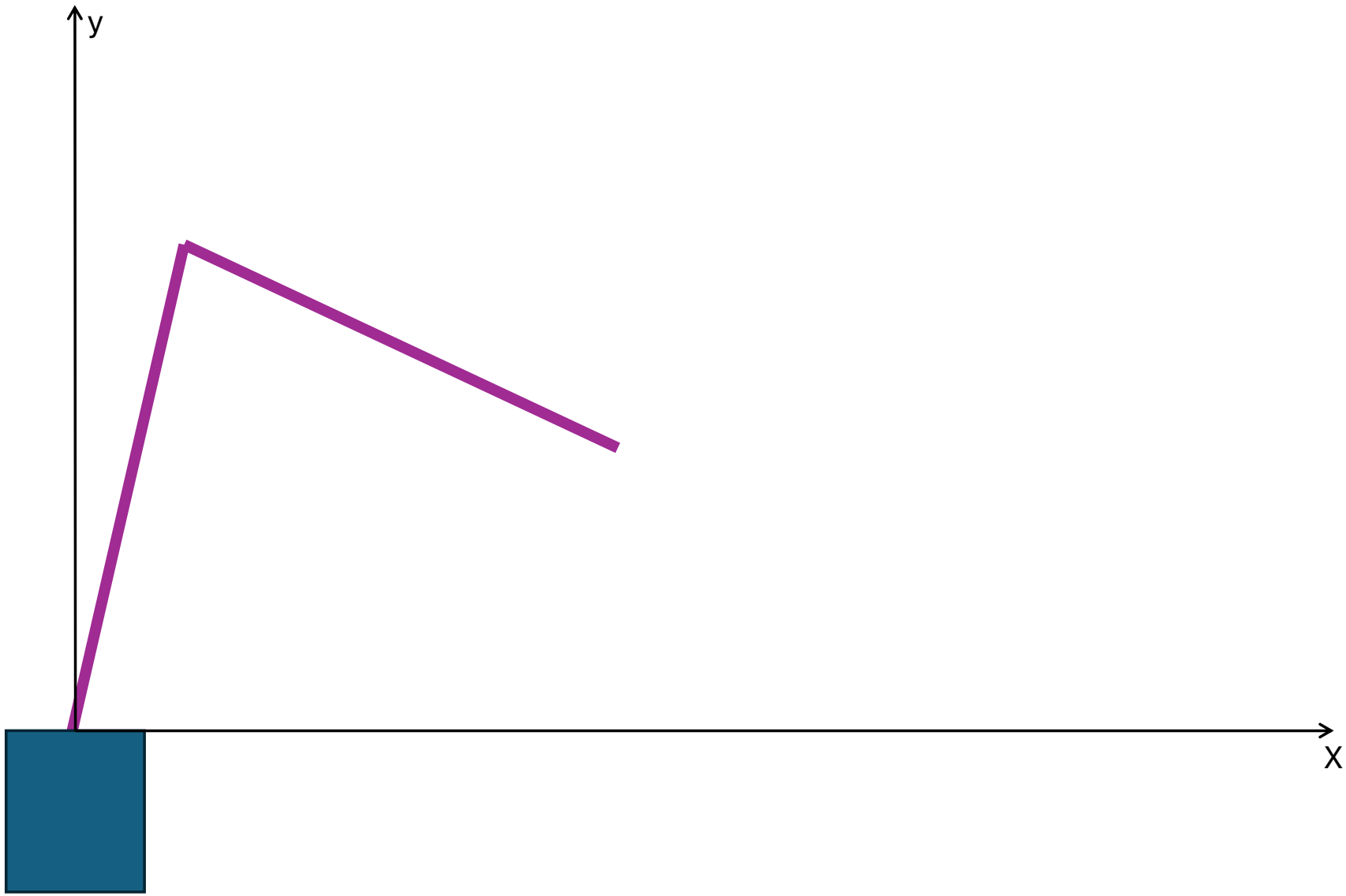
Forward Kinematics

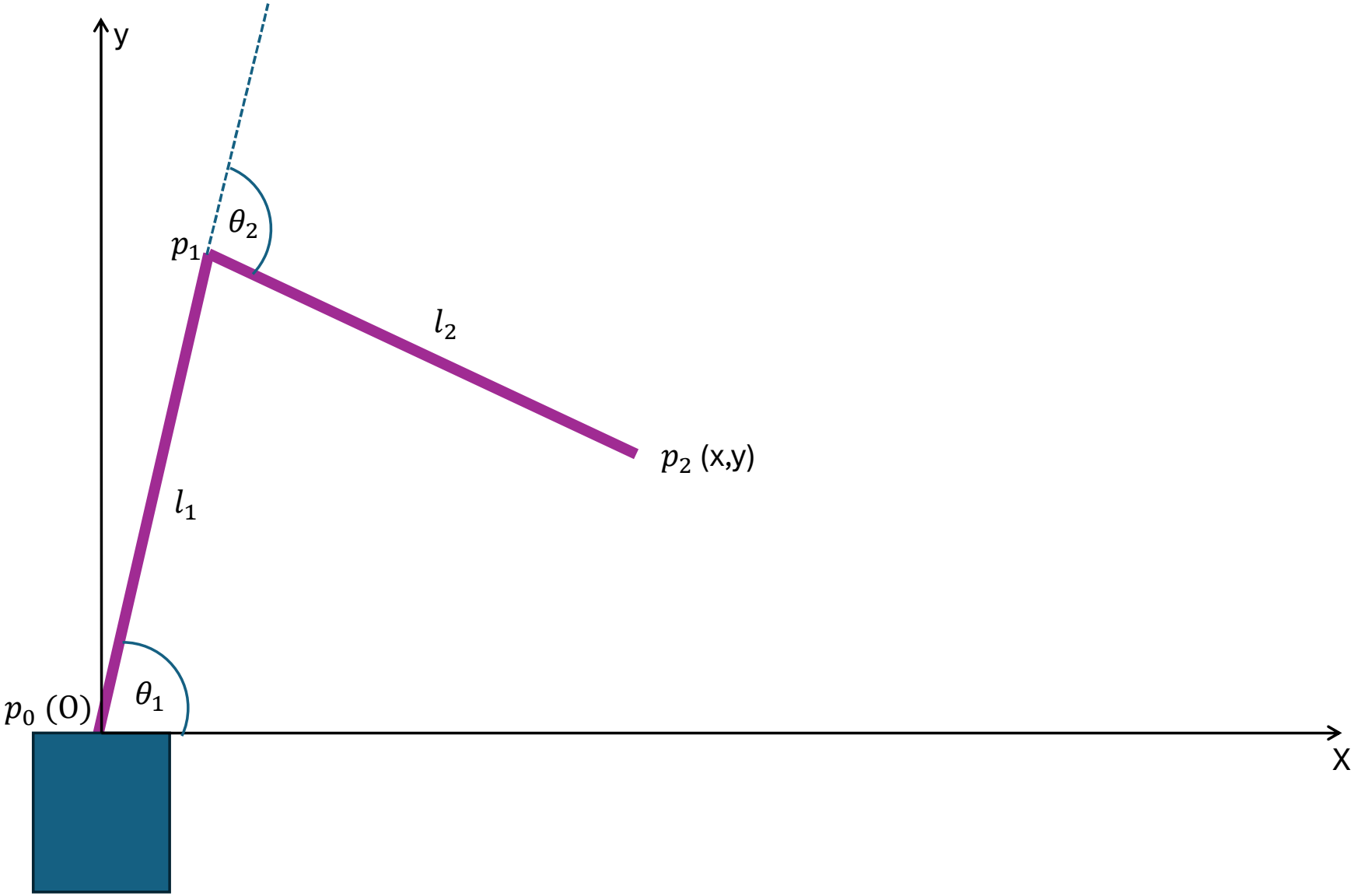
- “Where are we when we move our actuators?”



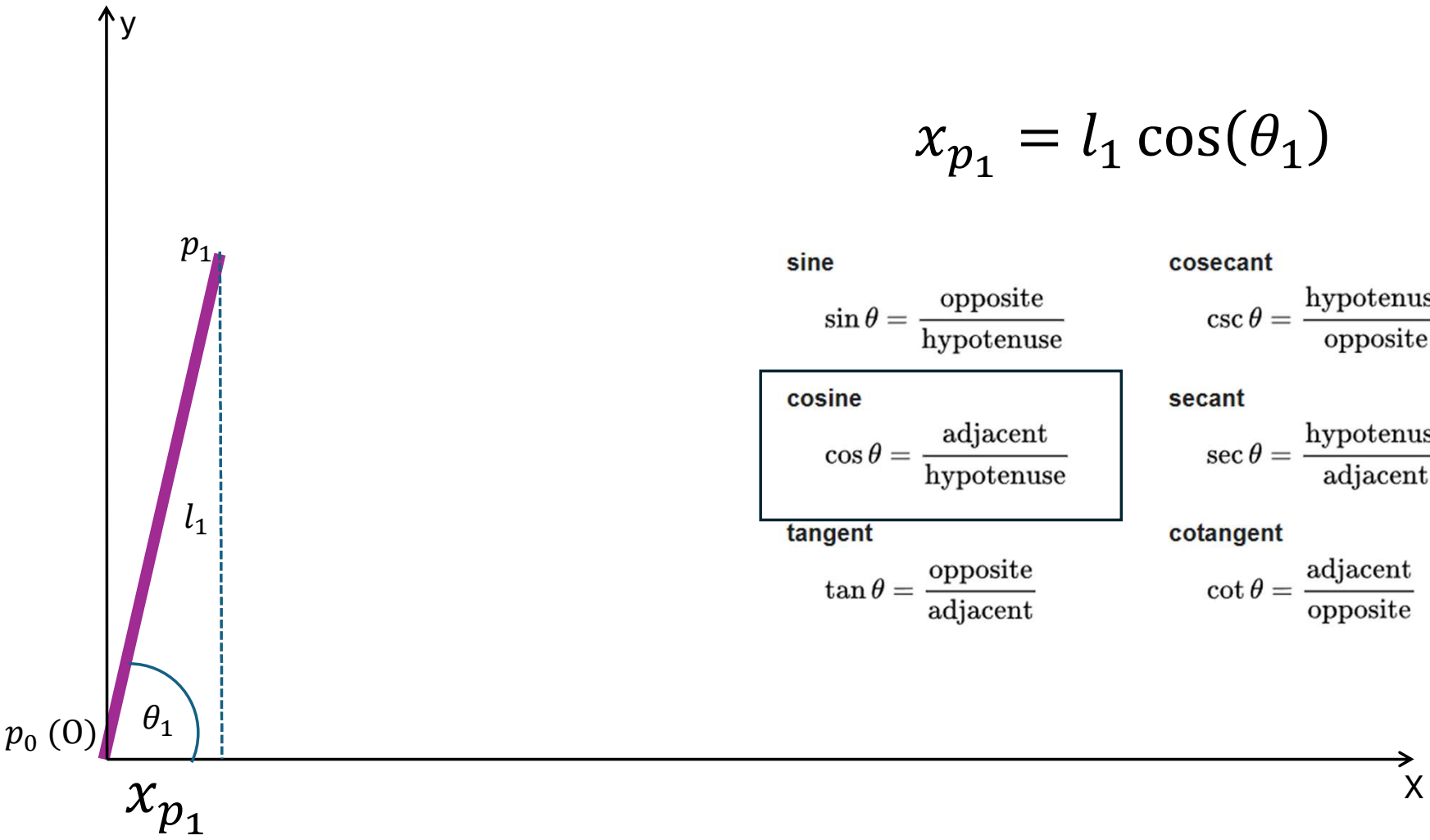





y
x


y
θ_2
p_1
l_2
p_2 (x,y)
l_1
θ_1
p_0 (O)
X

y
p_1
l_1
θ_1
p_0 (O)
x_{p_1}
X



$$x_{p_1} = l_1 \cos(\theta_1)$$

sine

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

cosecant

$$\csc\theta = \frac{\text{hypotenuse}}{\text{opposite}}$$

cosine

$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

secant

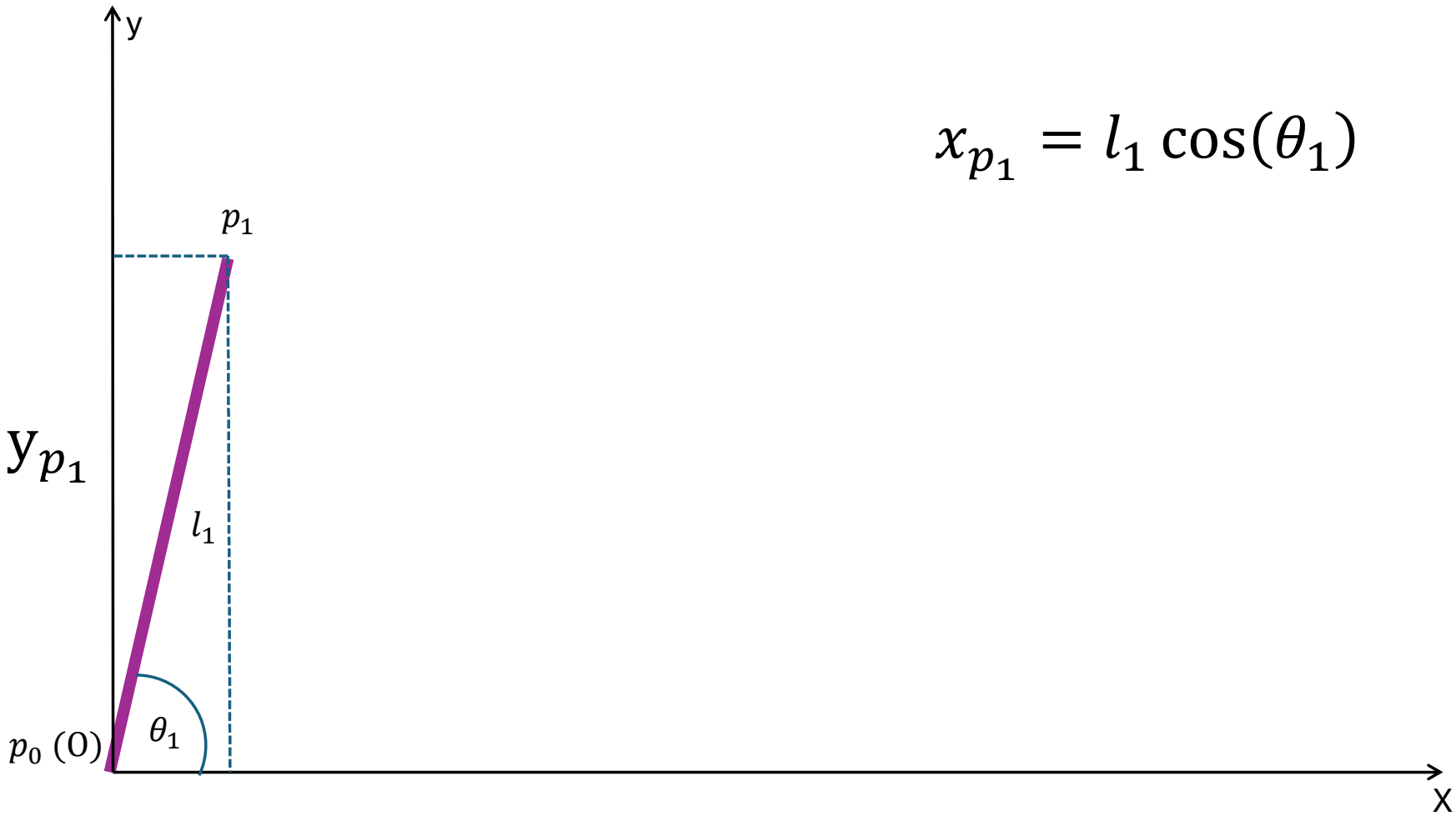
$$\sec\theta = \frac{\text{hypotenuse}}{\text{adjacent}}$$

tangent

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$$

cotangent

$$\cot\theta = \frac{\text{adjacent}}{\text{opposite}}$$



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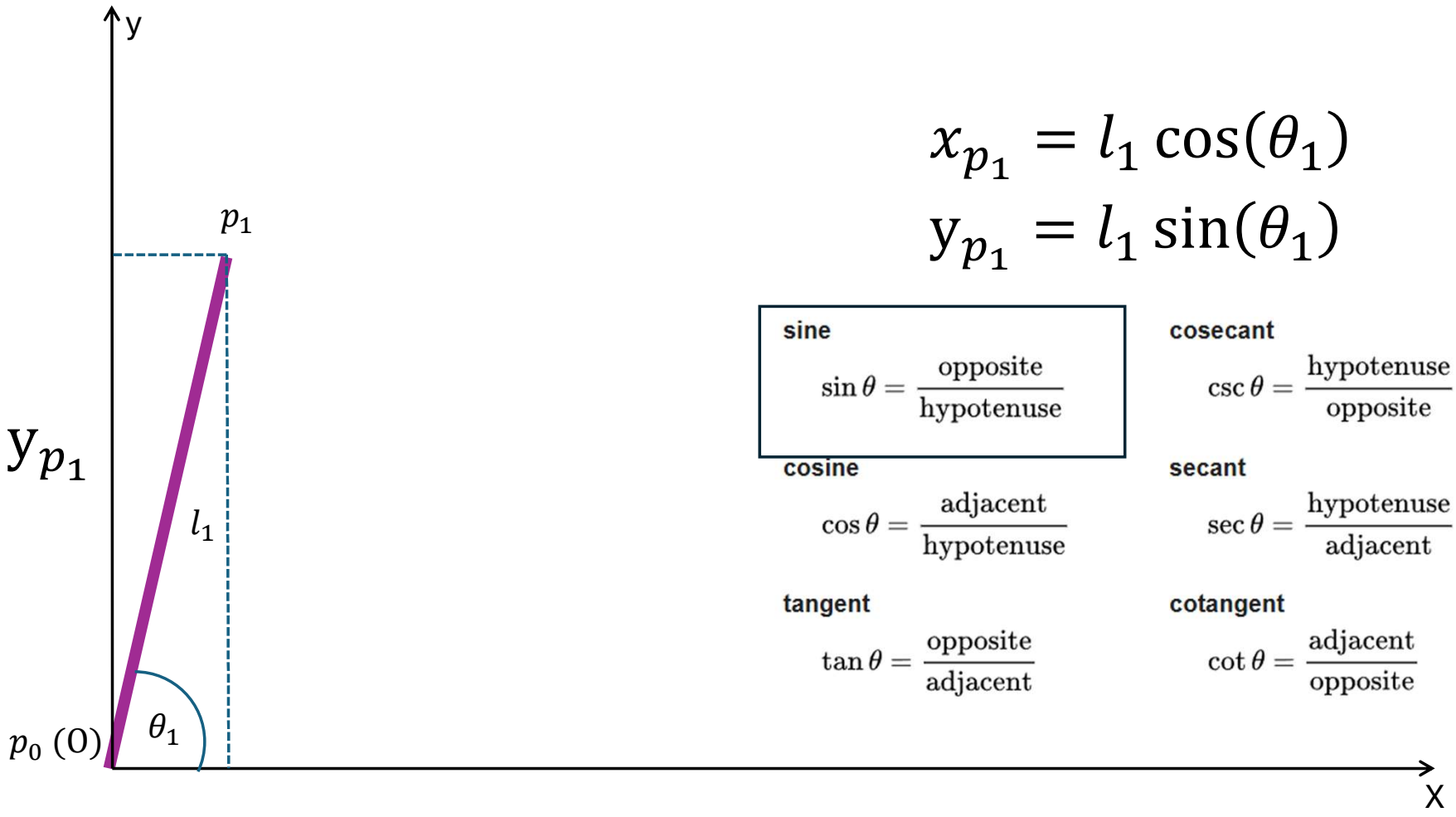


$$x_{p_1} = l_1 \cos(\theta_1)$$
$$y_{p_1} = l_1 \sin(\theta_1)$$

sine

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

cosecant

$$\csc\theta = \frac{\text{hypotenuse}}{\text{opposite}}$$

cosine

$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

secant

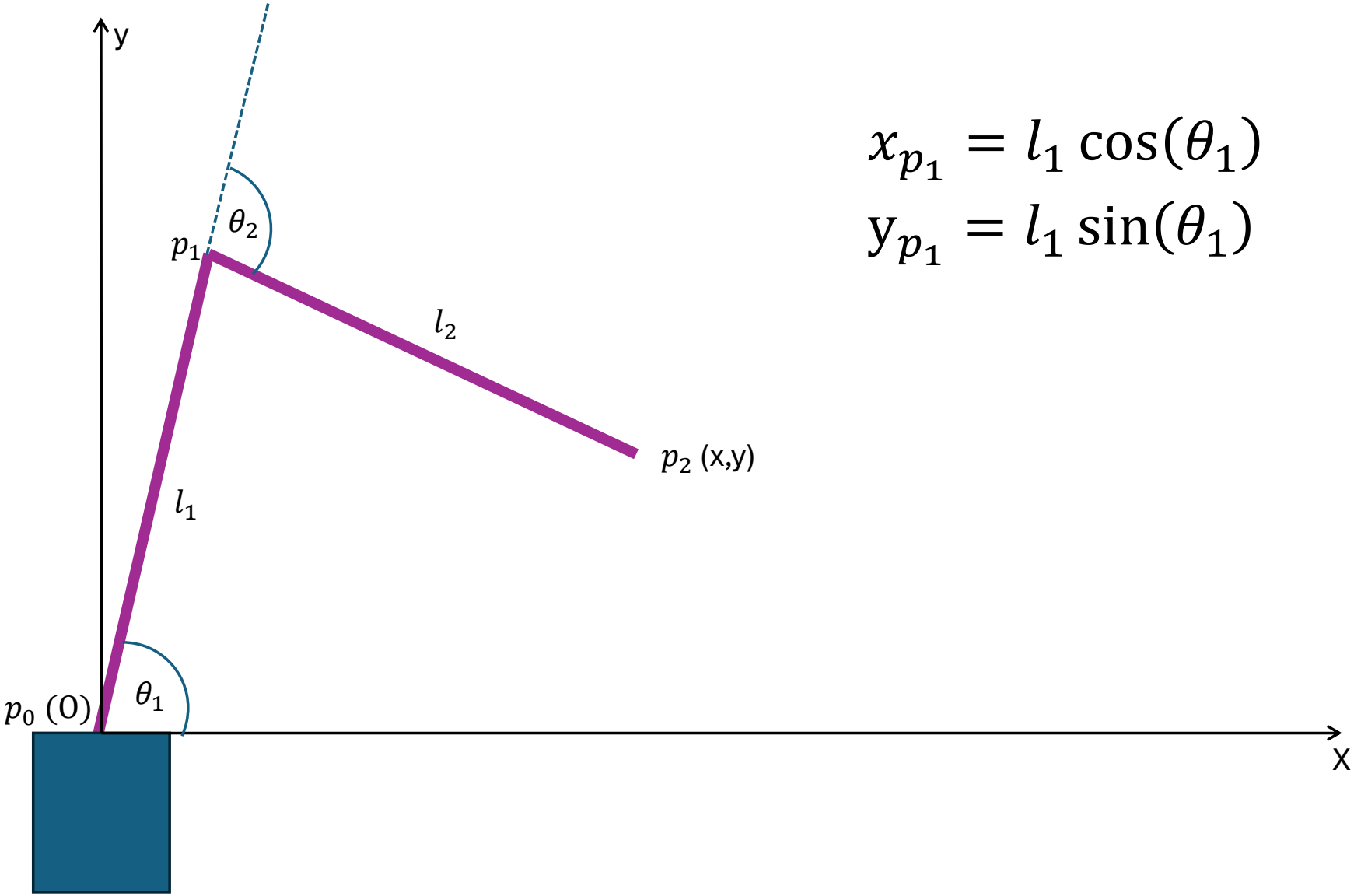
$$\sec\theta = \frac{\text{hypotenuse}}{\text{adjacent}}$$

tangent

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$$

cotangent

$$\cot\theta = \frac{\text{adjacent}}{\text{opposite}}$$



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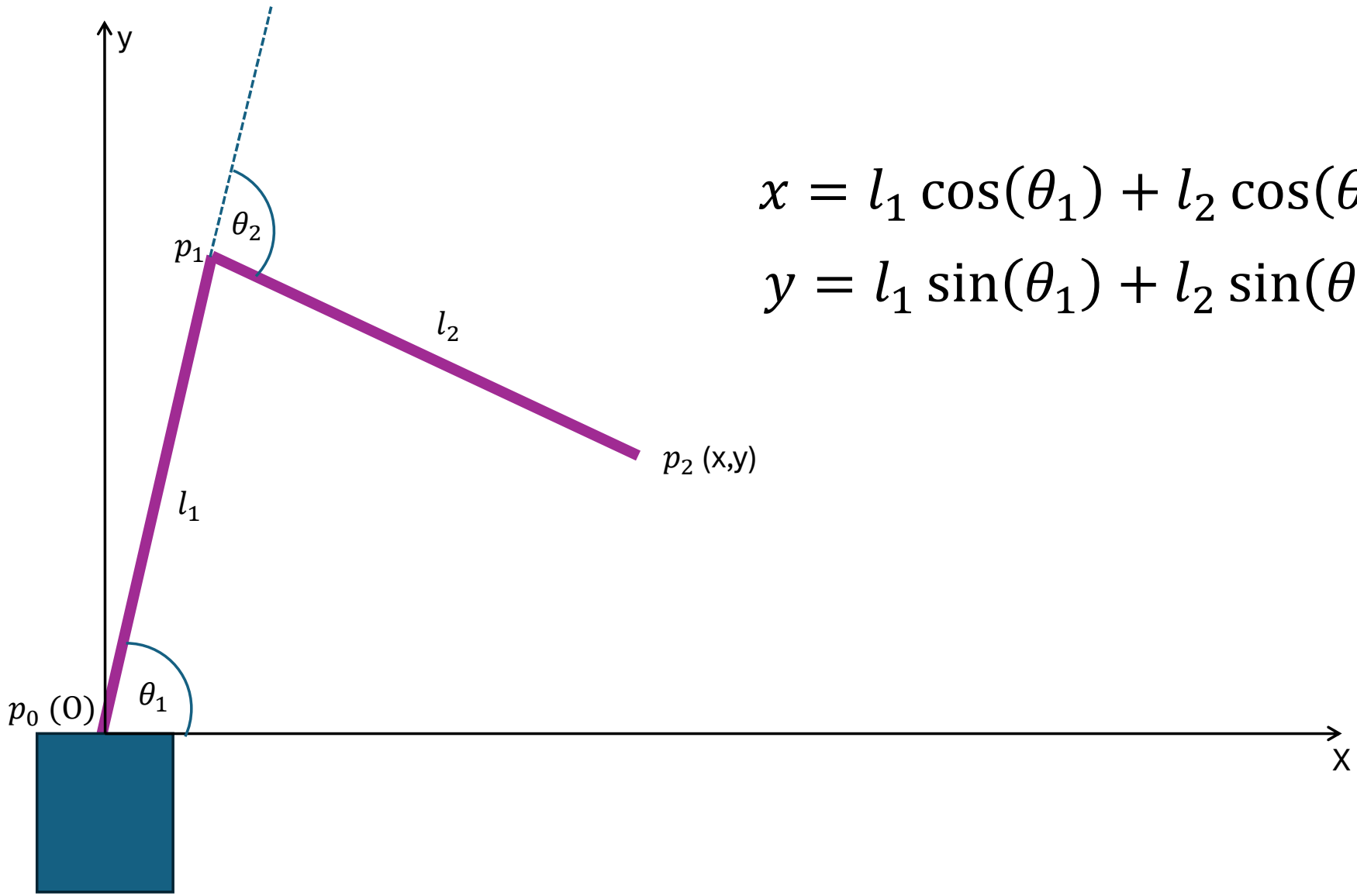
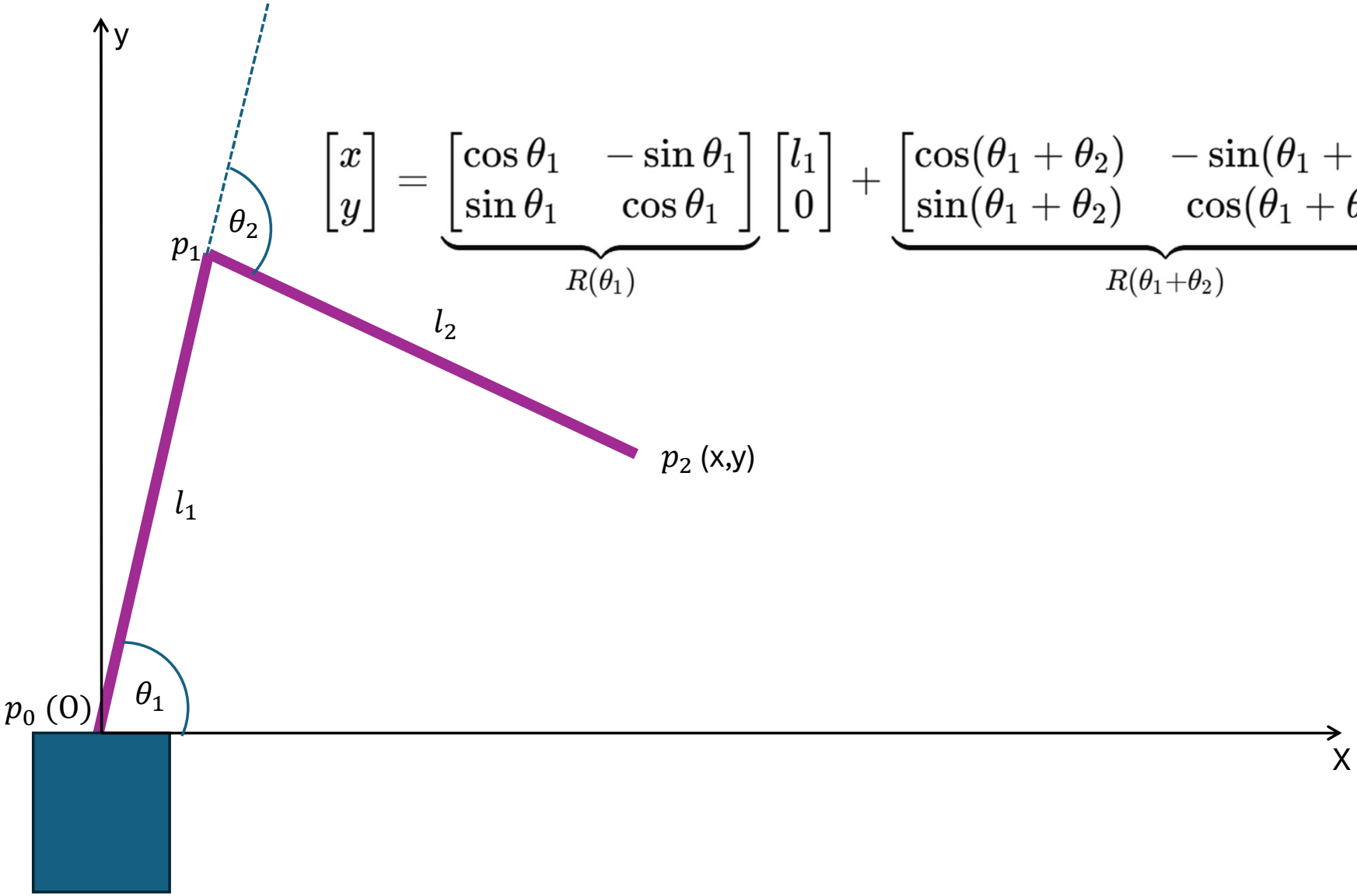
$$y_{p_1} = l_1 \sin(\theta_1)$$



$$x = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2)$$

$$y = l_1 \sin(\theta_1) + l_2 \sin(\theta_1 + \theta_2)$$



$$\begin{bmatrix} x \\ y \end{bmatrix} = \underbrace{\begin{bmatrix} \cos\theta_1 & -\sin\theta_1 \\ \sin\theta_1 & \cos\theta_1 \end{bmatrix}}_{R(\theta_1)} \begin{bmatrix} l_1 \\ 0 \end{bmatrix} + \underbrace{\begin{bmatrix} \cos(\theta_1+\theta_2) & -\sin(\theta_1+\theta_2) \\ \sin(\theta_1+\theta_2) & \cos(\theta_1+\theta_2) \end{bmatrix}}_{R(\theta_1+\theta_2)} \begin{bmatrix} l_2 \\ 0 \end{bmatrix}$$